

Electricity Price Forecasting Pipeline (PTF)

This project implements an **end-to-end machine learning pipeline** for forecasting the Turkish electricity market price (**PTF**) using data from the **EPIAŞ Transparency Platform**.

The goal of this project is not only to build a forecasting model, but to design a **reproducible, modular, and production-style data pipeline** that includes:

- automated data ingestion
- data preprocessing
- time-series feature engineering
- baseline and statistical model comparison
- machine learning modeling
- experiment tracking
- workflow orchestration

The pipeline uses:

- **LightGBM** for machine learning forecasting
- **MLflow** for experiment tracking
- **Apache Airflow** for workflow orchestration

System Architecture

The system is implemented as a **modular pipeline**, where each stage is an independent component.

flowchart TD

A[EPIAŞ Transparency API]

B[Data Ingestion

fetch_epias.py]

C[Raw Dataset

Parquet Storage]

D[Data Processing

process_epias.py]

E[Processed Dataset]

F[Feature Engineering

build_features.py]

G[Train Mode]

H[Model Training

LightGBM]

I[Model Evaluation]

J[MLflow Tracking]

K[Inference Mode]

L[Prediction]

```

A --> B
B --> C
C --> D
D --> E
E --> F

F --> G
G --> H
H --> I
I --> J

F --> K
K --> L

%% Styles
classDef api fill:#f8d7da,stroke:#333,stroke-width:2px,color:#000
classDef data fill:#d6eaf8,stroke:#333,stroke-width:2px,color:#000
classDef process fill:#d5f5e3,stroke:#333,stroke-width:2px,color:#000
classDef model fill:#fcf3cf,stroke:#333,stroke-width:2px,color:#000
classDef inference fill:#e8daef,stroke:#333,stroke-width:2px,color:#000

class A api
class C,E data
class B,D,F process
class H,I,J model
class G,K,L inference

```

Data Ingestion

Electricity price data is collected from the **EPIAŞ Transparency Platform API**.

The ingestion script performs the following steps:

- authentication through the **CAS authentication service**
- downloading **MCP / PTF market price data**
- transforming API responses into structured **Pandas DataFrames**
- storing the results as **partitioned Parquet datasets**

Incremental Data Fetching

The ingestion pipeline supports **incremental updates**.

Instead of downloading the entire dataset every time:

- the script checks the **latest timestamp** in the existing dataset
- data fetching starts **from that point**
- a small **overlap window** is used to avoid missing observations

This design allows the pipeline to run efficiently in **scheduled workflows**.

Data Processing

Raw API responses require cleaning before modeling.

The processing stage performs:

- column name standardization
- timestamp parsing
- numeric conversion of the `ptf` column
- corrupted record removal
- duplicate timestamp removal
- chronological sorting
- detection of missing hourly observations

The output is a **clean and structured time-series dataset** ready for feature engineering.

Exploratory Data Analysis

Initial analysis of the time series revealed several important patterns:

- strong **hourly structure**
- strong **daily seasonality**
- visible **weekly patterns**
- strong predictive power of lag features such as **lag_24** and **lag_168**

These observations guided the **feature engineering strategy and model design**.

Feature Engineering

Feature engineering is a **core component** of the pipeline.

The following feature groups are generated.

Time Features

- hour of day
- day of week
- weekend indicator
- cyclic transformations (**sin / cos encoding**)

Lag Features

- `lag_1`
- `lag_24`
- `lag_168`
- `lag_336`

Rolling Statistics

- rolling_mean_24
- rolling_mean_168

Difference Features

- diff_1
- diff_24
- diff_168

Train Mode

In **train mode**, the pipeline generates a target variable:

target = ptf shifted by 24 hours

This allows the model to **predict the electricity price at the same hour on the following day**.

Inference Mode

In **inference mode**:

- only **features are generated**
- the **target column is not created**

This allows the same pipeline to be used for **real-time prediction**.

Baseline Model

A simple baseline model was implemented using the **lag-24 approach**.

The prediction rule is:

$$PTF(t) \approx PTF(t - 24)$$

Electricity markets often exhibit **strong daily repetition**, making this baseline surprisingly competitive.

Model	MAE	RMSE	MAPE
Baseline (lag_24)	457.11	723.33	136.89

SARIMA Model

A classical statistical model was implemented for comparison.

Selected configuration:

SARIMA(2,1,2)(1,1,0,24)

The **seasonal period of 24 hours** captures the daily structure of electricity prices.

Model	MAE	RMSE	MAPE
SARIMA (2,1,2)(1,1,0,24)	558.27	863.37	200.75

The SARIMA model **did not outperform the lag-based baseline.**

LightGBM Model

A machine learning approach was implemented using **LightGBM**.

The dataset was split chronologically into:

- training set
- validation set
- test set

Best Hyperparameters

Parameter	Value
n_estimators	800
learning_rate	0.02
max_depth	6
num_leaves	31
min_child_samples	50
subsample	1.0
colsample_bytree	0.8

Validation Results

Metric	Value
MAE	296.55
RMSE	406.59
MAPE	69.41

Test Results

Metric	Value
MAE	392.49
RMSE	548.30
MAPE	246.74

Model Comparison

Model	MAE	RMSE	MAPE
Baseline (lag_24)	457.11	723.33	136.89
SARIMA	558.27	863.37	200.75
LightGBM	392.49	548.30	246.74

The **LightGBM model achieved the lowest MAE and RMSE**, indicating better predictive performance.

The large **MAPE values** are caused by periods where **PTF approaches zero**, which makes percentage-based errors unstable.

Experiment Tracking

The training pipeline integrates **MLflow** for experiment tracking.

MLflow logs the following information for each experiment:

- hyperparameters
- validation metrics
- test metrics
- trained model artifacts

This allows:

- reproducibility of experiments
- easy comparison between different model runs
- centralized storage of experiment results

Workflow Automation

The pipeline is orchestrated using **Apache Airflow**.

Two workflows were implemented:

- **Training Pipeline**
 - data ingestion
 - data processing
 - feature engineering
 - model training
 - model evaluation
 - MLflow logging
- **Inference Pipeline**
 - incremental data ingestion
 - data processing

- inference feature generation
- loading the trained model
- generating electricity price predictions

Retraining Pipeline

```

flowchart LR
    fetch_epias[fetch_epias]
    process_epias[process_epias]
    build_features[build_features]
    train_mode[train mode]
    train_lgbm[train_lgbm]
    evaluate[evaluate]

    fetch_epias --> process_epias
    process_epias --> build_features
    build_features --> train_lgbm
    train_lgbm --> evaluate

%% Styles
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classDef processing fill:#d5f5e3,stroke:#333,stroke-width:2px,color:#000
classDef training fill:#fcf3cf,stroke:#333,stroke-width:2px,color:#000
classDef evaluation fill:#f8d7da,stroke:#333,stroke-width:2px,color:#000

class fetch_epias ingestion
class process_epias processing
class build_features processing
class train_lgbm training
class evaluate evaluation

```

This workflow periodically retrains the forecasting model using newly available data.

Inference Pipeline

```

flowchart LR
    fetch_epias[fetch_epias]
    process_epias[process_epias]
    build_features[build_features]
    inference_mode[inference mode]
    predict[predict]

    fetch_epias --> process_epias
    process_epias --> build_features
    build_features --> predict

%% Styles
classDef ingestion fill:#d6eaf8,stroke:#333,stroke-width:2px,color:#000
classDef processing fill:#d5f5e3,stroke:#333,stroke-width:2px,color:#000
classDef inference fill:#e8daef,stroke:#333,stroke-width:2px,color:#000

```

```
class fetch_epias ingestion
class process_epias processing
class build_features processing
class predict inference
```

This workflow generates new price predictions using the latest data.

Separating training and inference pipelines improves reliability and maintainability.

Technologies Used

The project uses the following technologies and tools:

- **Python**
- **Pandas**
- **LightGBM**
- **Statsmodels**
- **MLflow**
- **Apache Airflow**
- **Parquet**
- **EPIAŞ Transparency Platform API**

Conclusion

This project demonstrates the design of a **production-style machine learning pipeline** for electricity price forecasting.

The system includes:

- automated data ingestion
- robust data preprocessing
- time-series feature engineering
- baseline and statistical model comparison
- machine learning modeling
- experiment tracking with **MLflow**
- workflow orchestration using **Apache Airflow**

The final architecture is **modular**, **reproducible**, and suitable for **automated retraining and prediction generation**.